MATH 213 Homework 12

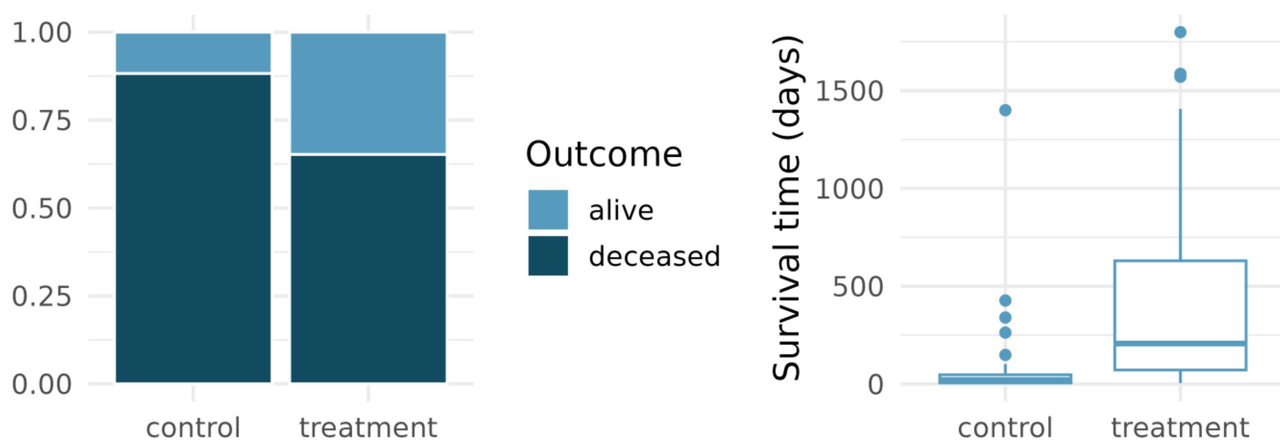
Reading: We have been working on material from Chapter 11 of our textbook. Please skim through that chapter to give you another chance to push these ideas through your head. Note that the book discusses cutoffs for when to consider a p-value "small enough" to consider a result to be unusual, which I haven't yet done in class. We'll talk about that alpha value more next week.

Written homework:

Complete Question 8 in the Chapter 11 exercises. Two notes:

1. One part asks about box plots, which I skipped over in class, but you can read about them in section 5.5 of the book. It's a visual way of representing data along with medians and quartiles that's quite useful. I'll answer questions in class.
2. Part d. ii. Is the only time I intend to ask you this much detail about what underlies this "randomization test" procedure, but I did want you to walk through the idea one more time on your own.

Heart transplants. The Stanford University Heart Transplant Study was conducted to determine whether an experimental heart transplant program increased lifespan. Each patient entering the program was designated an official heart transplant candidate, meaning that they were gravely ill and would likely benefit from a new heart. Some patients got a transplant and some did not. The variable `transplant` indicates which group the patients were in; patients in the treatment group got a transplant and those in the control group did not. Of the 34 patients in the control group, 30 died. Of the 69 people in the treatment group, 45 died. Another variable called `survived` was used to indicate whether the patient was alive at the end of the study.[11] (Turnbull, Brown, and Hu 1974)



a. Does the stacked bar plot indicate that survival is independent of whether the patient got a transplant? Explain your reasoning.
   No, there is a difference in survival rates between patients who receive a transplant and those who did not, suggesting that survival is not independent of transplant status in the control group a larger proportion of the bar is dedicated to patients who died(deceased) indicating a higher mortality rate. However, the treatment groups show a much larger proportion of survivors indicting a better survival outcome.

b. What do the box plots suggest about the efficacy of heart transplants.
The plots show that patients in the treatment group (who received a transplant) had a higher median survival time, more variability and some long surviving outliers compared to the control group, which had a much lower median and shorter survival times overall. This suggests that heart transplants generally lead to longer survival times.

c. What proportions of patients in the treatment and control groups died?

Control: 30/34 = 0.882 * 100 = 88.2%

Treatment: 45/69 = 0.652 * 100 = 65.2%

d. One approach for investigating whether the treatment is discernably effective is randomization testing.
    a. What are the claims being tested?
    Null hypothesis: survival is independent of treatment. The transplant does not affect survival.

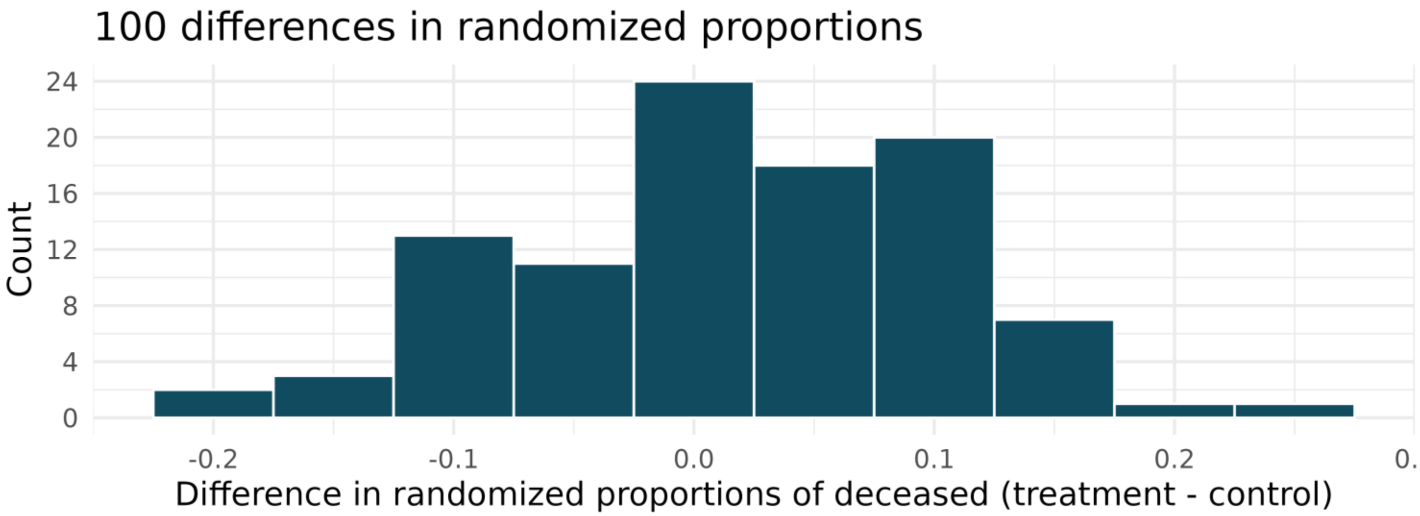
    Alternative hypothesis: survival depends on the treatment. The transplant improves survival chances.

    b. The paragraph below describes the set up for a randomization test, if we were to do it without using statistical software. Fill in the blanks with a number or phrase.

We write *alive* on 28 cards representing patients who were alive at the end of the study, and *deceased* on 75 cards representing patients who were not. Then, we shuffle these cards and split them into two groups: one group of size 69 representing treatment, and another group of size 34 representing control. We calculate the difference between the proportion of cards in the treatment and control groups (treatment - control) and record this value. We repeat this 100 times to build a distribution centered at 0. Lastly, we calculate the proportion of simulations where the simulated differences in proportions are greater than or equal to the observed difference. If this proportion is low, we conclude that it is unlikely to have observed such an outcome by chance and that the null hypothesis should be rejected in favor of the alternative.

    c. What do the simulation results shown below suggest about the effectiveness of heart transplants?

The results below who that the observed difference in death rates between the treatment and control groups is much larger the many difference that occurred in the 100 random simulations. Since a difference as extreme as the observed one was very unlikely under the assumption of no treatment effect, this suggests heart transplants significantly reduce mortality. Therefore, they are effective



100 differences in randomized proportions
Count
24
20
16
12
8
4
0
-0.2
-0.1
0.0
0.1
0.2
0.3
Difference in randomized proportions of deceased (treatment - control)